

Candidate Discovery & Ranking AI

A High-Performance Staged Screening System for Senior AI Engineers

Participant Team: Antigravity

Compute Target: CPU Only, $\leq$ 16GB RAM

Target Role: Senior AI Engineer (Founding Team)

Execution Time: ~1 min 55 sec

The Challenge & Dataset Traps

Recruiting platforms are flooded with keyword-heavy profiles, but matching requires deep logic. Our solution addresses several adversarial traps deliberately built into the challenge pool:

- **Honeypots (~80 profiles):** Synthetic profiles with mathematically impossible properties, such as 8 years of experience at a company founded 3 years ago, or expert proficiency in multiple skills with 0 years used. If honeypot rate exceeds 10% in the top 100, the submission is disqualified.
- **Keyword Stuffers:** Candidates who list dozens of advanced AI/ML skills on their profile but have never deployed or used them in their career history descriptions.
- **Consulting-Only Profiles:** General services candidates (TCS, Infosys, etc.) with keyword matches but zero product-engineering exposure or startup alignment.

System Architecture Overview

We built a multi-stage deterministic screening pipeline that processes candidates locally on CPU within the 5-minute limit:

Stage 1: Hard Filtering	Excludes the 70 dynamically detected honeypots and 9,739 consulting-only candidates before scoring.
Stage 2: Hybrid Scoring	Scores the remaining 90,191 valid candidates on Skills (45%), Experience (30%), Behavioral Signals (15%), and Fit (10%).
Stage 3: Deterministic Sort	Orders candidates by Score descending, breaking ties using Candidate ID ascending to ensure perfect reproducibility.
Stage 4: Reasoning Generation	Composes specific, non-templated 1-2 sentence rationales referencing actual experience, locations, and notice periods.

Deep Dive: Honeypot Detection Rules

We designed a dynamic check that caught exactly **70 honeypots** in the 100K pool, avoiding hardcoding for generalized capability:

1. **Impossible Job Duration:** Compares job start/end dates against the stated duration. E.g., CAND_0007353 has a job starting in 2023 with duration of 166 months (impossible elapsed time).
2. **YoE vs Job History Mismatch:** Calculates sum of durations in career history and flags if it differs from profile YoE by > 4.0 years. E.g. claiming 13 YoE but having only 11 months of career history.
3. **Expert Skill 0 Duration:** Detects candidates listing 3 or more expert/advanced skills but with 0 months used.
4. **Impossible Skill Duration:** Flags candidates claiming skill durations exceeding their total YoE (e.g. claiming 92 months of RAG experience with only 3 years of total experience).

Deep Dive: Staged Scoring Model

Scoring components are weighted to align with a Series-A product engineering founding role:

Skills Score (45%)	Matches core skills (embeddings, vector DBs, Python, ranking eval). Crucial: applies a 65% discount if the skill term is not found in the candidate's job descriptions, neutralizing keyword stuffers.
Experience Score (30%)	Targets the 5-9 YoE sweet spot. Inspects career text for active engineering action verbs (<i>shipped, deployed, scaled, infrastructure</i>) and rewards Tier-1 education.
Behavioral Score (15%)	Evaluates platform availability and responsiveness. Penalizes response rates < 15% and login inactivity > 6 months. Rewards active GitHub profiles.
Logistics & Fit (10%)	Scores location preferences (Noida/Pune preferred, relocatable Tier-1) and notice period (buyout required for 90+ days).

Results & Compliance

The system was run locally and evaluated under strict constraints:

- **Execution speed:** The complete run on the 100K pool took **1 minute 55 seconds** on a standard CPU, far below the 5-minute budget.
- **Memory and Network:** Fully self-contained. No external API calls are made and memory usage is kept minimal (< 1.5 GB).
- **Honeypot Rate:** 0% honeypot rate in the top 100 shortlist, satisfying the strict disqualification check.
- **Reasoning Quality:** Custom sentence generation produces rationales referencing specific companies (e.g. Zomato, Meta, Google), locations, notice periods, and honest warnings (e.g. 90-day notice buyout required). No templates or repeats are used.

Validator Verdict:

Submission is valid.